

SQL QUERY

DATA ANALYTICS (PROJECT 3)

DATASET FROM [DECOLAB](#)

EXECUTIVE SUMMARY

Confidential This report presents a detailed analysis of 1,200 e-commerce orders extracted from the DECOLAB database using structured SQL queries. The dataset spans January 2023 through mid-2025, covering seven product categories, five payment methods, and five customer acquisition channels. Each section includes the exact SQL query used, the result set, and data-driven insights.

KEY KPIS AT A GLANCE

Total Revenue	Total Orders	Avg Order Value	Units Sold
\$1,264,762	1,200	\$1,054	3,535 Per Order
USD	Transactions	Units	Item

Insight: Despite healthy total revenue of \$1.26M, only 19.2% of orders were successfully Delivered. Nearly half (41.4%) were either Returned (20.6%) or Cancelled (20.8%), signaling critical fulfilment gaps requiring immediate operational attention.

SECTION 1 — DATASET OVERVIEW & SCHEMA

The first SQL query retrieves the full dataset to confirm structure, column types, and a preview of records before deeper analysis begins.

SQL Query — Dataset Preview — First 10 Records

```
SELECT TOP 10 *  
FROM [dbo].[Dataset for Data Analytics (Project 3)]
```

<u>OrderId</u>	<u>Date</u>	<u>CustomerID</u>	<u>Product</u>	<u>Quantity</u>
ORD200000	1/4/2023	C72649	Monitor	5
ORD200001	8/23/2024	C75739	Phone	2
ORD200002	2/27/2024	C81728	Tablet	5
ORD200003	10/15/2023	C33540	Chair	1
ORD200004	5/8/2025	C81840	Printer	4
ORD200005	10/23/2023	C37249	Phone	2
ORD200006	6/17/2025	C83492	Laptop	1
ORD200007	5/12/2023	C41460	Monitor	5
ORD200008	4/2/2025	C26817	Phone	2

SQL Query — Dataset Exploration

```
SELECT DISTINCT Product  
FROM [dbo].[Dataset for Data Analytics (Project 3)]  
  
SELECT DISTINCT PaymentMethod  
FROM [dbo].[Dataset for Data Analytics (Project 3)]  
  
SELECT DISTINCT CouponCode  
FROM [dbo].[Dataset for Data Analytics (Project 3)]  
  
SELECT DISTINCT ReferralSource
```

Statistics	Result
Coupon Code	WINTER 15, FREESHIP, NO COUPON CODE, SAVE10
Product	Monitor, Phone, Tablet, Chair, Laptop, Desktop, Printer.
Payment Method	Online, Debit Card, Gift Card, Cash, Credit Card.
Referral Source	Google, Facebook, Referral, Email, Instagram

SECTION 2 — KPI ANALYSIS CORE BUSINESS

KPIs are computed using SQL aggregation functions: SUM, AVG, and COUNT.

SQL Query — Core KPIs — Revenue, AOV, Units Sold

```
SELECT COUNT(*) AS TotalOrders
FROM [dbo].[Dataset for Data Analytics (Project 3)]
SELECT SUM (TotalPrice) AS TotalRevenue
FROM [dbo].[Dataset for Data Analytics (Project 3)]
SELECT AVG(TotalPrice) AS AverageOrderValue
FROM [dbo].[Dataset for Data Analytics (Project 3)]
SELECT SUM(Quantity) As TotalUnitsSold
FROM [dbo].[Dataset for Data Analytics (Project 3)]
```

KPI's	Result
Total Orders	1200
Total Revenue	1,264,761.96
Average Order Value	1,053.97
Total unit Sold	3,535.00

SQL Query — Order Count

```
SELECT OrderStatus, Count(*) AS Orders
FROM [dbo].[Dataset for Data Analytics (Project 3)]
GROUP BY OrderStatus;
```

Order Status	Order
Pending	237
Returned	247
Delivered	231
Cancelled	250
Shipped	235

SQL Query — Revenue by Payment Method

```
SELECT PaymentMethod, SUM(TotalPrice) AS Revenue
FROM [dbo].[Dataset for Data Analytics (Project 3)]
Group By PaymentMethod
Order By Revenue DESC;
```

Payment Method	Revenue
Credit Card	263,874.63
Online	262,442.94
Cash	259,786.29
Gift Card	246,323.92
Debit Card	232,361.18

SECTION 3 — PRODUCT PERFORMANCE & REFERRAL CHANNEL

SQL Query — Units Sold by Product Category

```
SELECT [Product],  
SUM(Quantity) As QuantitySold  
from [dbo].[Dataset for Data Analytics (Project 3)]  
Group By [Product] Order by QuantitySold DESC;
```

Product	Quantity Sold
Chair	562
Printer	542
Laptop	535
Desk	508
Tablet	497
Monitor	480
Phone	411

SQL Query — Revenue by Product Category

```
SELECT [Product],  
SUM(TotalPrice) As Revenue  
FROM [dbo].[Dataset for Data Analytics (Project 3)]  
GROUP BY [Product] ORDER BY Revenue DESC;
```

Product	Revenue
Chair	195,620.1
Printer	195,612.6
Laptop	192,126.6
Desk	186,569
Tablet	175,651.4

Monitor	167,459.9
Phone	151,722.4

SQL Query — Orders & Revenue by Referral Source

```
SELECT ReferralSource,
COUNT(*) AS Orders
FROM [dbo].[Dataset for Data Analytics (Project 3)]
GROUP BY ReferralSource ORDER BY Orders DESC;
```

Referral Source	Order
Instagram	259
Email	250
Google	241
Facebook	228
Referral	222

Insight: Chair and Printer lead both revenue and volume rankings. Phone is the weakest performer — lowest revenue (\$151.7K) and fewest units sold (411). A dedicated Phone promotion or pricing review is recommended.

Instagram is the channel by both volume (259 orders), and Referral word-of-mouth is the weakest — the referral program needs stronger financial incentives to compete with paid channels.

SECTION 4 — REVENUE TREND & SEASONALITY

SQL Query — Monthly Revenue

```
SELECT MONTH(Date) AS SalesMonth,  
SUM(TotalPrice) AS Revenue  
FROM [dbo].[Dataset for Data Analytics (Project 3)]  
GROUP BY MONTH(Date) ORDER BY SalesMonth;
```

Months	Revenue
January	124313.2299
February	112344.7799
March	123840.9297
April	109186.0508
May	135142.5903
June	170616.1305
July	85784.63997
August	86343.20989
September	69321.65033
October	89834.82056
November	75493.42975
December	82540.50016

SQL Query — Monthly Order Count

```
SELECT MONTH(Date) As SalesMonth,  
COUNT (*) AS Orders  
FROM [dbo].[Dataset for Data Analytics (Project 3)]  
GROUP BY MONTH(Date) ORDER BY SalesMonth;
```

Months	Order
January	106
February	106
March	128
April	113
May	120
June	147
July	87
August	79
September	73
October	78
November	76
December	87

Insight: June is the peak month (\$170.6K, 147 orders). September is the weakest (\$69.3K, 73 orders). Q3 (Jul-Sep) consistently underperforms — a key window for targeted campaigns.